

Stellar PSFs Fake Hosts in Little Red Dots

Broad-band point-spread functions misassign light when source spectra differ from field stars.

arXiv:2608.30459 — source-conditioned polychromatic PSFs for compact high-redshift nuclei

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Background

Compact high-redshift sources are modelled by subtracting a point-spread function built from field stars. This assumes the target and the stars share the same spectral shape across the filter.

No one had quantified how much broad-band colour mismatch between source and stellar PSF reference biases inferred host flux.

What this paper is selling

The authors derive the physical and mathematical origin of this light-misassignment error and demonstrate a correction. Applied to little red dots, it revises sizes, luminosities and masses inferred from broad-band imaging.

Key results to remember

- A pure point source with no host still yields a spurious Sérsic component in F444W.
- Every tested nuclear spectrum, stellar reference and detector position exceeds the published 11.8% host fraction.
- The bias is systematic, not random, and is correctable for other compact distant sources.

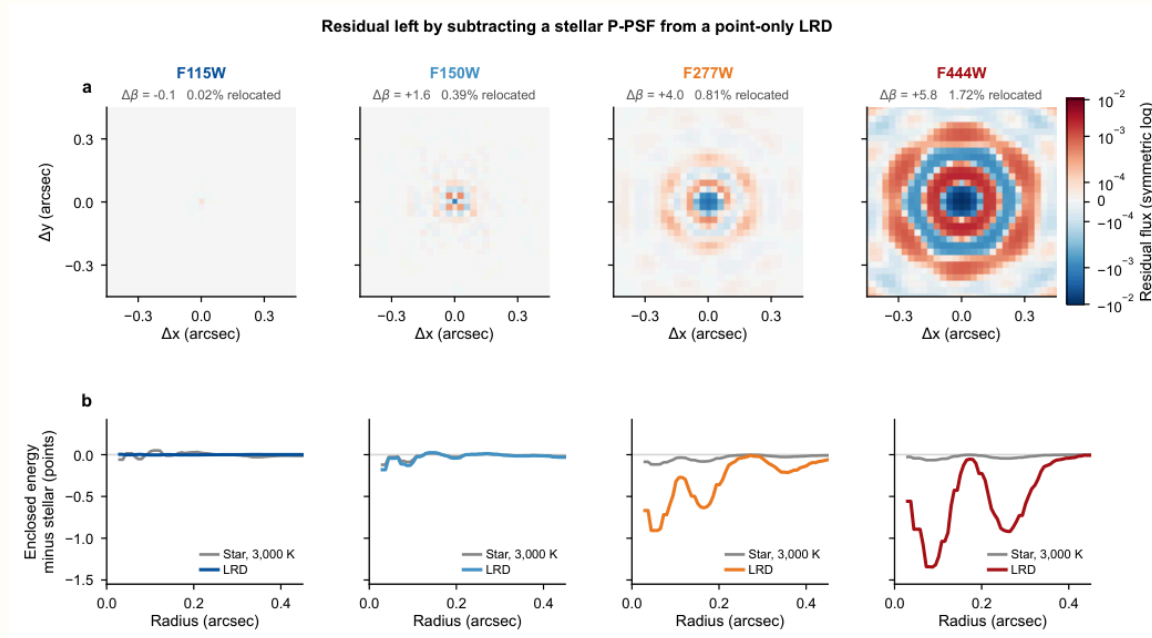


Figure 1 | Subtracting a stellar PSF from a point-only LRD leaves wavelength-dependent residuals that masquerade as a Sérsic host.

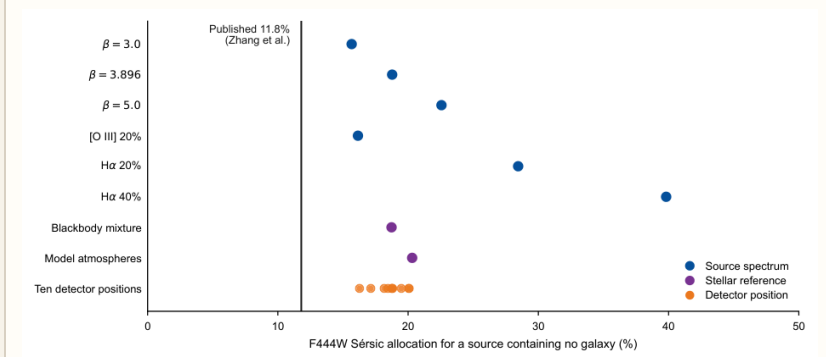


Figure 2 | Across all tested spectra, stellar references and detector positions, the spurious F444W host allocation stays above the published value.